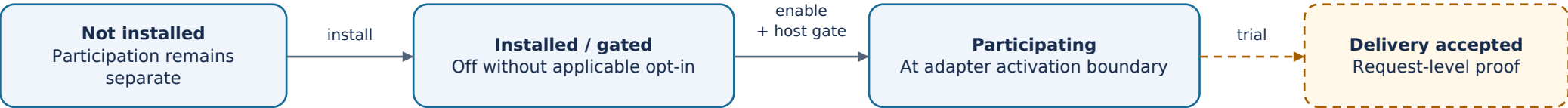
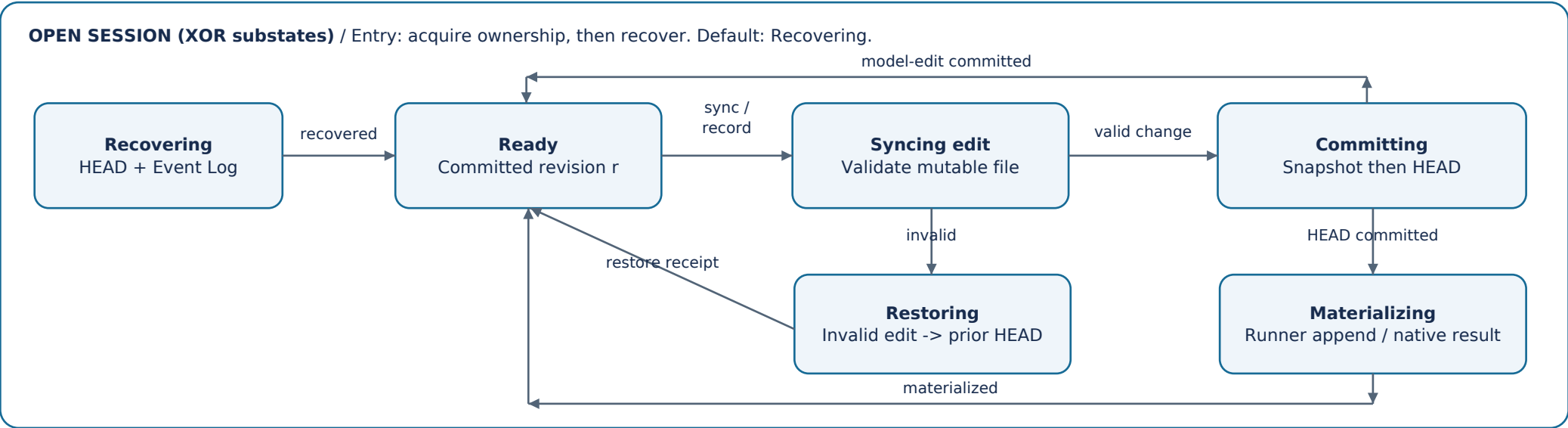


Activation and durable session lifecycle

Solid blue = published code or documented boundary. Dashed amber = required acceptance condition, not verified behavior. Labels use event [guard] / action. These are conceptual states mapped to existing functions, not an executable state-machine proposal.



Disable: write scoped off record; Claude applies on a new session, Codex rechecks each hook. **Uninstall:** remove this runner's owned config, preserve unrelated edits and retained session records. Installing alone never proves participation or delivery. Shared state/participation can activate both already-installed adapters.



record(): sync first -> append Event Log -> apply -> commit -> materialize. Model-edit commit skips materialization.
Unchanged sync returns Ready; invalid input without HEAD stays uncommitted (revision 0).
Crash before HEAD: remove orphan snapshot; after durable log: replay seq > HEAD.through; after HEAD/before file: rematerialize.

Outside the composite: live foreign lock -> Refused; session.close -> Closed (release in finally); reopen -> Recovering, never control-state history. Inactive adapters leave the host native. Core failure / ambiguous Claude boundary may stand aside rather than retry an uncertain append.

Implementation: [core/session.ts](#); [core/lock.ts](#); [core/participation.ts](#); [setup/install.ts](#).

Delivery boundaries are different

Editing context is not the same event as committing it, accepting a reset, or observing the next request. Keep those acknowledgments separate in release evidence.

CLAUDE DEFAULT | Full Replacement per user turn; Injection within a turn



Over budget -> native compaction -> record native result -> Compaction-only frame. Core fault / ambiguous frame / unsafe Bash-tail observation -> stand aside; plugin compaction may skip, explicit/manual native path remains. Main-session tool starts wait behind the boundary; external editors/subagents are not coordinated.

CLAUDE PER-STEP EXPERIMENT | Explicit opt-in; build+sync+fit -> opaque authorization handle -> custom send. Build/fit/auth/send failure delegates to the host. A host fallback is not proof of replacement. Keep this experiment off for initial acceptance.

CODEX PLUGIN | Full Replacement at agent-initiated resets; history grows between resets



Gate refusal preserves the current window. PreCompact token-limit/manual backstop has a readability/16 MiB bound, but no token-budget gate; label Compaction-only. A successful hook return cannot prove the agent read every part or what the next model request contains.

CODEX TURN LOOP (separate headless opt-in) | sync -> fresh ephemeral thread -> inject committed user message -> turn/start -> completion -> record. Plugin and token-budget reset path are disabled in its owned app-server. Revision 0 starts a fresh thread without injection.

Cancellation / recovery: timeout or uncertain start -> recover known turn ID -> interrupt -> wait for terminal completion. If uncertain, close owned RPC process and mark loop unavailable. close() stops admission, closes RPC, awaits in-flight recording, then releases session lock. A refused delivery has mode=null and no model call; no automatic old-thread continuation.

Sources: [Claude register](#); [Claude contract](#); [Codex hooks](#); [Codex turn loop](#).

Transition, invariant and test map

I = intended invariant; current code may violate it. P = proposed validation, not executed here. Filenames identify suites to extend, not proof of exact-case coverage.

Transition / source	Invariant and observable outcome	Minimal proposed trace / test home
T1 Off -> Participating participation(), locallyEnabled()	I1 No opt-in means no recording/import side effects. Kill switch wins; nearest explicit off blocks ancestor on.	P1 Parent on -> child off -> corrupt child record -> hook: remain inactive or error, never inherit on. participation.test.ts; codex-hook.test.ts
T2 Ready -> Sync -> Commit/Restore Core.sync(), record()	I2 Valid edits commit before appends; invalid edits restore and emit a receipt. Delivery uses the committed reply, not a later mutable-file reread.	P2 Edit old sentinel to new -> record -> sync: old absent from delivered revision. Invalid UTF-8/empty/linked/oversize -> restore. session/read/confinement tests
T3 Log -> Snapshot -> HEAD -> File Core.commit(), recover()	I3 HEAD is authority; each logged sequence is applied once. Cache loss changes cost, not truth. Process-crash safety and power-loss durability are distinct.	P3 Fault after log, before HEAD, before file; reopen -> one copy of each event; preserve intervening stale edit in log. Separate directory-fsync/power-loss test. recovery.test.ts
T4 Open/operate/close acquireLock(), serialized(), close()	I4 One owner; same-owner operations serialized; foreign live owner refused. Exit/error must not strand a lock. No guarantee of FIFO admission.	P4 Two contenders + owner crash; parallel hooks; close log error; retained-close A -> session B -> end B -> retry A. lock.test.ts; Claude hooks tests
T5 Claude boundary -> delivery/fallback session.compact, turn.step	I5 No duplicate tail recording after uncertainty; pinned prefix/permissions preserved; Working Context stays user authority. Mode describes actual boundary.	P5 Active tool or ambiguous frame -> stand aside; lost record reply -> no replay; oversized fallback summary -> preserve native result. Fresh session resets turn-local flags. Claude hooks/per-step tests
T6 Codex reset -> agent read-back resetRefusal(), PreCompact	I6 Agent reset denied if unsafe or over budget; native backstop only checks deliverability. Reset acceptance alone is not delivery proof.	P6 9 MiB context -> serialized CLI reply fits IPC or fails explicitly; restored receipt -> reset denied; change multipart digest -> restart read. codex-hook/read tests
T7 Fresh thread -> running -> interrupted/closed runOneTurn(), startAndWait()	I7 Never continue old history after failed injection. Correlate thread/turn IDs. Unknown activity cannot be reused; owned resources eventually released.	P7 Start succeeds -> inject fails -> repeat: no leaked subscription. Lost start reply + late completion; timeout/interrupt race; close during record. turn-loop.test.ts; jsonrpc.test.ts
T8 Disable/uninstall/rollback setup project + ledger	I8 Preserve unowned config, exact bytes, and retained session data. All private reads/writes stay confined before any external effect.	P8 Concurrent config edit + install failure -> preserve bytes; linked backup/ledger refused; absent managed file restored to absence where owned. setup hosted/review regression tests

Execution profile: synchronous core effects inside admitted calls; separate operation and append leases. Claude async boundary gate guards replacement but has no universal host-event queue guarantee. Codex notifications filter IDs and buffer early arrivals. No assumed child-first priority, global run-to-completion, orthogonal atomicity, history restoration, or model checker proof.

Release checks tied to counterexamples

The following are unresolved hosted review claims, with published-source observations where inspected. Counterexample sequences are test plans, not reproduced failures. Existing owner fixes may already change them; validate against the next published exact heads.

Review / target	Counterexample to validate	Required outcome / invariant
Nearest participation record	Ancestor on -> descendant off record malformed -> lookup ascends. Published findRecord()/locallyEnabled() visibly continue.	Fail closed at existing invalid nearest record (I1).
Revision reads; Event Log links	Replace revision/log with link -> open/sync or append. Published snapshot uses readBytes/readFileSync; log open/truncate are pathname-based.	No target bytes read, truncated, appended or delivered; verify descriptors and anchors (I8).
Directory creation; Backup writes; Ledger reads	Swap managed parent or backup tree -> mkdir/copy/read. Post-check must not come after external mutation.	No out-of-root effect; confined create/read/write throughout (I8).
Snapshot durability	Snapshot rename -> HEAD rename -> power loss -> HEAD survives but snapshot entry does not.	Fsync appropriate parent directories; test actual durability assumptions (I3).
Whole-log reads; Cited source size; Hook IPC	Large log/source or duplicated 9 MiB reply -> allocation/IPC bound exceeded before small output limit.	Bound/stream inputs; size IPC for serialized payload, including escaping (I2/I6).
Retained closes; Turn-local flags	Close A fails -> start B -> close B; or A edits -> new session -> first Read.	Keep cleanup ownership; clear session-local stubbing flags (I4/I5).
Fallback budget; Per-step cost log	Fallback summary still over budget -> frame it; credential-like synthetic payload -> heuristic log path.	Do not frame undeliverable summary; no durable credential payload copies (I5/I8).
Failed thread injection	thread/start succeeds -> inject rejects -> retry. Published catch precedes assigning threadId/cleanup.	Dispose fresh thread; preserve prior handle without continuing it (I7).
Marker-only repair; Escaped TOML; Rollback bytes; Rollback absence	Keep markers but remove values; escaped equivalent key; invalid bytes decode equally; created tracked file fails install.	Validate effective owned config; byte-exact concurrent-edit checks; restore owned absence (I8).

Evidence and release gate. Claude PR #1 reports 29 passed / 11 explicit Codex-only skips (40); Codex PR #1 reports 63 passed / 4 Claude-only skips + 1 real-host hold (68), plus pinned typechecks. These are author-reported affected setup runs, not rerun here. Prior full suites belong to prior heads. Real installation, next-request delivery and long-session acceptance remain unverified.

Scope. Verified published heads: Claude e5da333fa144caf041feceb0fb1bbff120f6326c; Codex 644d02b313d79017cfece32a0ddf066b8a940c3f. Both SOURCE.json manifests declare the same 30 core hashes and core patch 783b50984a37c01ad527d31a099d41737b15c526. Owner-reported local candidates 9409e65 / 8e878d1 and core 7a50fab were unavailable and are not represented as verified fixes. Next: exact-head targeted validation + independent review + Jeff approval; supported-host acceptance remains separate.